Higher Moments, Highly Matter:

Optimal Monetary Policy in an Uncertain World

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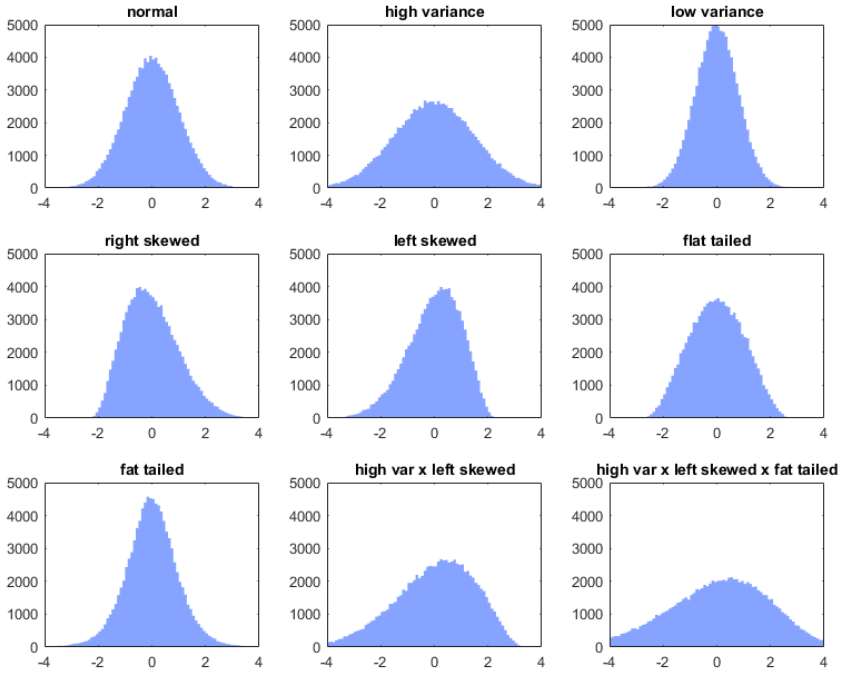
* The views expressed herein are those of the author, and do not necessarily represent those of the Bank of Thailand

** I am grateful for all fruitful advice and comments from Asst. Prof. Dr. Pongsak Luangaram. All errors are on my own.

Statistical Moments & Standard Academic Models



Source: Author's calculation

- Standard monetary policy model consists of
  1) Central Bank's Objective: Quadratic Loss
  2) Macro model: Linear
  3) Uncertainty/Shock: Additive
  - Monetary policy rule does not respond to uncertainty.
  - Ex. Taylor Rule (1999)

$$i_t = 2 + \pi_t + 0.5(\pi_t - \pi^*) + (y_t - y_t^*)$$

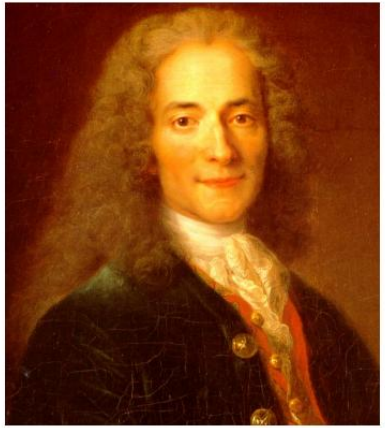
- This result is known as **certainty equivalence**.

Certainty Equivalence in an Uncertain World

Uncertainty Outside and Inside Economic Models*

Prize Lecture, December 8, 2013

by Lars Peter Hansen
University of Chicago, Chicago, IL, USA.



Le doute n'est pas une condition agréable, mais la certitude est absurde. Voltaire (1776)

Translation: *Doubt is unpleasant but certainty is absurd.*

This paper is a theoretical study about…

How should monetary policy be conducted under uncertainty?

- **Revisit the role of monetary policy under uncertainty, by focusing on the tail risk rather than on standard stabilization policy.**

Central Bank's Objective functions	Functional forms	Role of Monetary Policy
Quadratic loss (2nd moment)	$L = (y - \bar{y})^2 + \lambda(\pi - \pi^*)^2$	Macro stabilization
Quartic loss (4th moment)	$L = (y - \bar{y})^4 + \gamma(\pi - \pi^*)^4$	Minimizing tail risks

- Thus far, GaR approach (proposed by the IMF) is gaining popularity among central banks but studies so far are mostly empirical without theoretical justification.
- **This paper aims to provide a theoretical insight of how monetary policy should respond to tail risk.**

The Bank of England's Approach:

Towards the Grand Unifying Theory of Monetary Policy and Macroprudential Policy

Monetary Policy vs Macroprudential Policy

Monetary policymaker's loss function

$$\min_{\{r_{t+i}\}_{i=0}^{T}} \mathcal{L}_t \equiv E_t \left\{ \sum_{i=o}^{H} \beta^i [(\pi_{t+i} - \pi^*)^2 + \lambda (y_{t+i} - y_{t+i}^*)^2] \right\}$$

Policy Rate:

Stabilize macroeconomy

Separate function

Macroprudential policymaker's loss function

$$\min_{\rho_t} \mathcal{L} \equiv E_t \left\{ \sum_{i=0}^{T} \beta^i \left[f(G@R_{t+i}) - \phi y_{t+i} \right] \right\}$$

Macroprudential Policy:

Minimize Growth at Risk

Source: Carney (2020)

Related Literature

Tinbergen-Theil Framework (1950's)	Extension
Type of Uncertainty	**Baseline: Additive Uncertainty** Parameter Uncertainty (Brainard, 1967; Soderstrom, 2002; Kimura and Kurozumi, 2003) Model Uncertainty (Levin, Wieland and Williams, 1999; Levin and William, 2003) Measurement Uncertainty (Wieland , 1998; Orphanides, 2002; Swanson, 2004)
Structure of the economy	**Baseline: Linear Structure** Zonelinear Philips Curve (Orphanides and Wieland, 1999) Convex Philips Curve (Schaling, 2004) Effective Lower Bound (Lengwiler, 1998; Chung et al. ,2011)
Policy Tools	**Baseline: Policy Rate** Forward Guidance (Fermia et al., 2013)
Central Bank Loss Function	**Baseline: Quadratic Loss Function with Two Objectives** Inflation Zone Targeting (Boinet and Martinn, 2005; Nowaihi and Stracca, 2001)

Setup: How to model uncertainty

- This study is based on IS, PC and monetary policy rule.

- An economy is divided into two phases.
  - In the first phase, the policy maker perceives first round shocks and chooses the degree and direction of response to stabilize the economy.
  - In the second phase, the economy is hit by second round shock after the policy was decided. **Hence, the authority cannot fully control the final outcome of the economy.**

- Calibration: $\alpha = 0.082^{*}$ $\beta = 0.25^{**}$ $\omega = 0.7^{***}$

* based on Goodhart (2001) and transformed by Okun's law. ** and *** based on Blanchard (2015)

Second-round shock

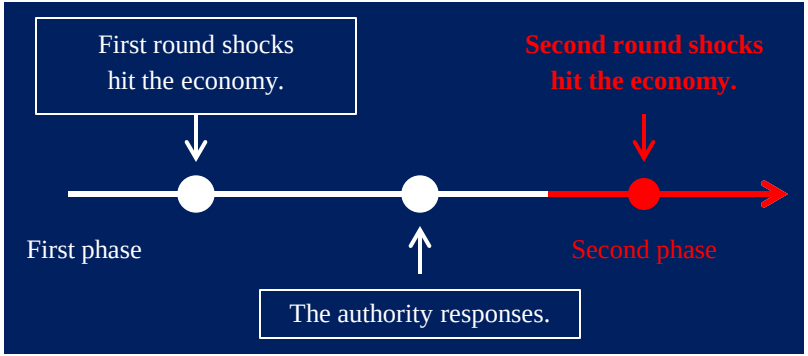
$$U = U_n + \alpha(i - \pi^e - r*) + \varepsilon + \boxed{\varepsilon^s} \quad (1)$$

$$\pi = \pi^e - \beta(U - U_n) + \nu + \boxed{\nu^s} \quad (2)$$

$$\pi^e = (1 - \omega)\pi_{-1} + \omega\pi* \quad (3)$$

$$U_{CB} = U_n + \alpha(i - \pi^e - r^*) + \varepsilon \quad (4)$$

$$\pi_{CB} = \pi^e - \beta(U - U_n) + \nu \quad (5)$$



Solving for Monetary Policy Rule (Certainty Case)

- Following Clouse (2018) **this study employed quartic loss function** while the quadratic loss function is used as a baseline.

$$L = (U - U_n)^2 + \lambda(\pi - \pi^*)^2$$

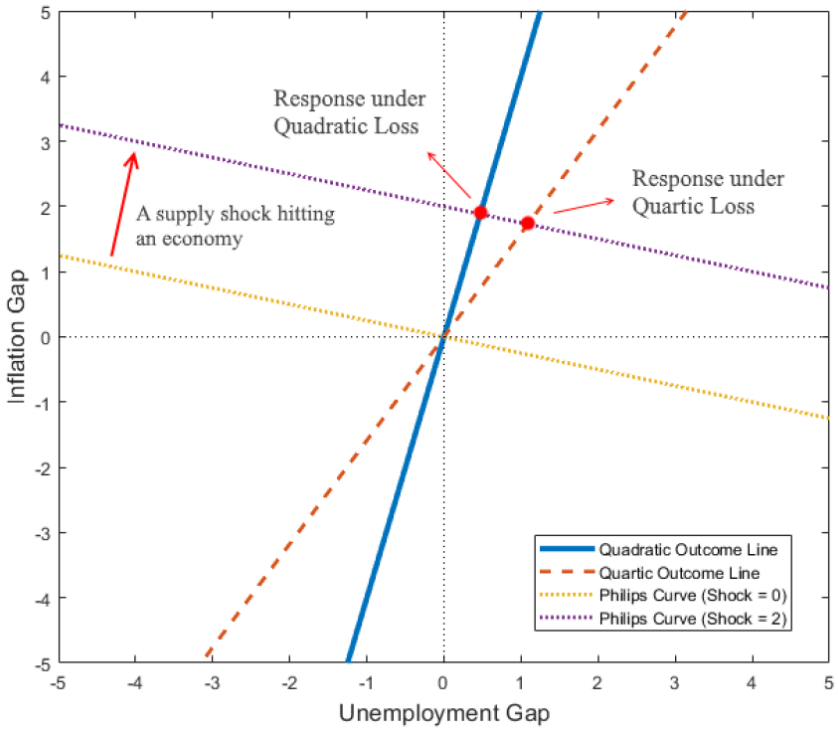
$$L = (U - U_n)^4 + \gamma(\pi - \pi^*)^4$$

- Solving for first order condition, we obtain the first order conditions constituting MPRs.

$$U_{CB} - U_n = \lambda\beta(\pi_{CB} - \pi^*)$$

and

$$(U_{CB} - U_n)^3 = \gamma\beta(\pi_{CB} - \pi^*)^3$$



Source: Author's calculation

Solving for Monetary Policy Rule (Uncertainty Case)

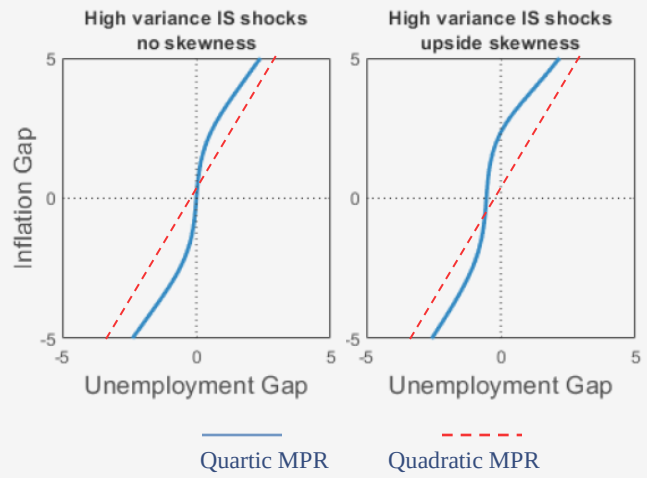
- In presence of uncertainty, the central bank <u>minimizes expected loss function.</u>

$$E(L) = (U_{CB} - U_n)^2 + \lambda(\pi_{CB} - \pi^*)^2 + \lambda(1 + \beta^2)\sigma_{\nu'}^2 + \sigma_{\varepsilon'}^2$$

and

$$E(L) = (U_{CB} - U_n)^4 + 6(U_{CB} - U_n)^2\sigma_{\varepsilon^s}^2 + 4(U_{CB} - U_n)\sigma_{\varepsilon^s}^3 S_{\varepsilon^s} + \sigma_{\varepsilon^s}^4 K_{\varepsilon^s}$$

$$+\gamma((\pi_{CB} - \pi^*)^4 + 6(\pi_{CB} - \pi^*)^2\sigma_{\nu^s}^2 + 4(\pi_{CB} - \pi^*)\sigma_{\nu^s}^3 S_{\nu^s} + \sigma_{\nu^s}^4 K_{\nu^s})$$



Source: Clouse (2018) and modified by Author

Taking first differentiation, Clouse (2018) obtained

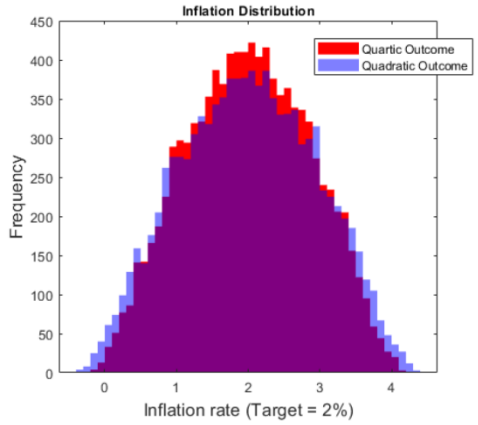
$$U_{CB} - U_n = \lambda\beta(\pi_{CB} - \pi^*)$$

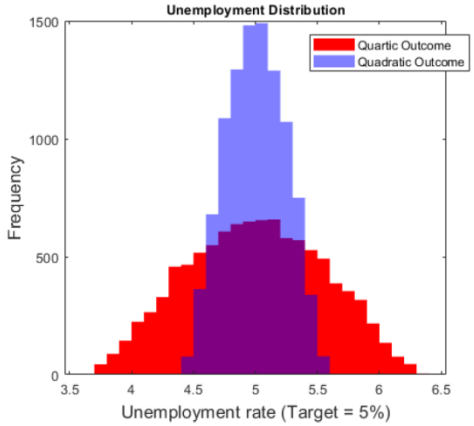
and $$(U_{CB} - U_n)^3 + 3(U_{CB} - U_n)\underline{\sigma_{\varepsilon^s}^2} + \underline{\sigma_{\varepsilon^s}^3 S_{\varepsilon^s}} = \gamma\beta((\pi_{CB} - \pi^*)^3 + 3(\pi_{CB} - \pi^*)\underline{\sigma_{\nu^s}^2} + \underline{\sigma_{\nu^s}^3 S_{\nu^s}})$$

- Note that now the optimal response in quartic case depends on variance and skewness of both shocks.

Properties of the Quartic Loss Function

Results from Monte Carlo Simulation





Source: Author's calculation

At first glance, optimal monetary policy under quartic loss function seems to heavily stabilize inflation…

Is a central banker with quartic loss function more hawkish?

Answer: No, we cannot judge the stance of the central bank just by comparing the distributional outcome since any distribution can be a part of the whole possible outcome.

Hence, we need some other tools to evaluate both monetary policies in a more holistic way.

Evaluating Monetary Policy using Taylor Curve

- Second moment Philips curve or Taylor curve is one way to evaluate the performance of monetary policy, by depicting the trade-off in variance of output and inflation.

- Q: The question is how does the Taylor curve of quartic solutions look like?

Possibility 1: If the quartic function generates points **below** the benchmark, then the quadratic does not truly minimize the loss.

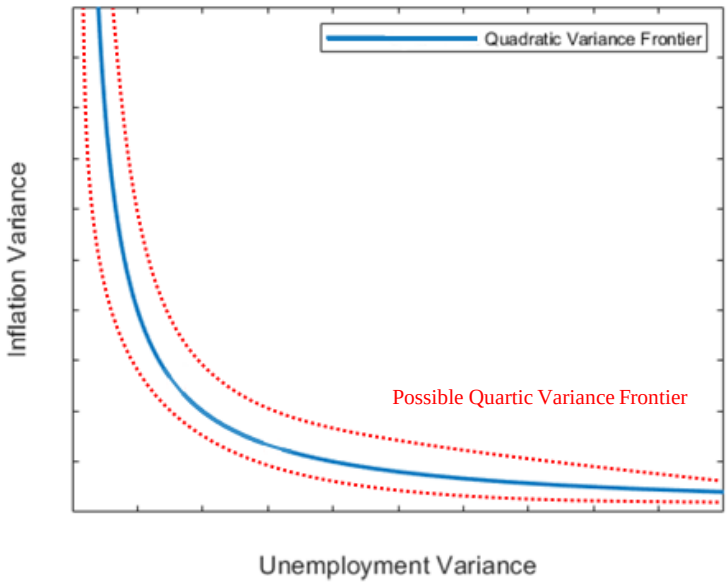
Hence, this contradicts the optimization problem.

Possibility 2: If the quartic function generates the points **above** the benchmark, then the quartic does not truly minimize the loss.

Hence, this contradicts the optimization problem.

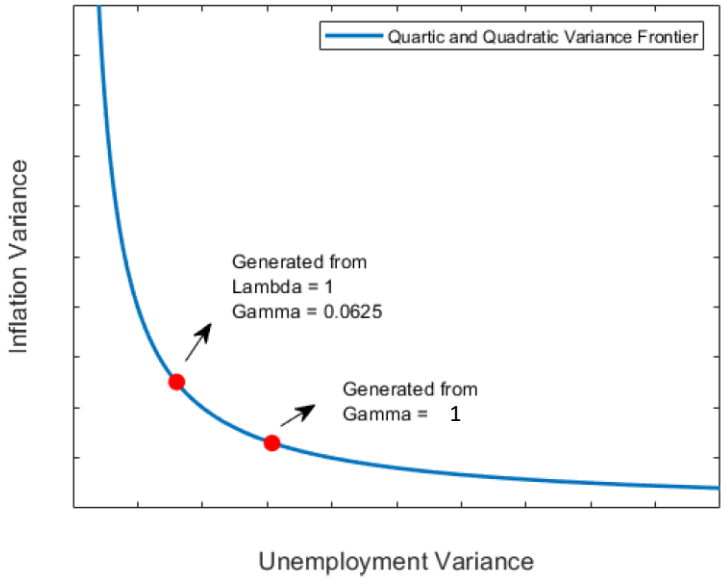
$$L = (U - U_n)^2 + \lambda(\pi - \pi^*)^2$$

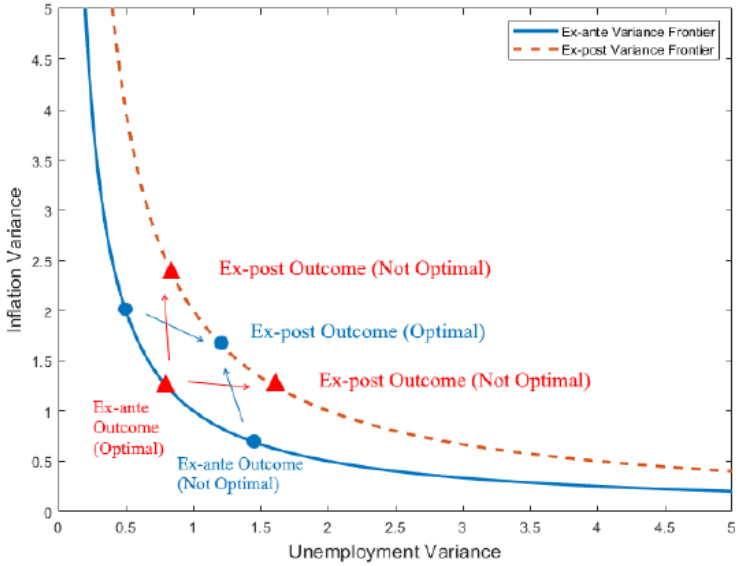
$$L = (U - U_n)^4 + \gamma(\pi - \pi^*)^4$$



Ex-post Taylor curve

- Hence, **the two loss functions produce the same Taylor Curve**, where each pint is generated from the different weights. However, the quartic loss function aims to minimize the ex-post loss rather than the ex-ante loss.





Source: Author's calculation

Linking Two Losses: Gamma Transformation

- Since the two loss functions generate the same set of solutions, there may be a linkage bridging the two specification together. Luckily, there exists a way to understand the connection between quartic loss function and quadratic loss function.

- **There must be some value of lambda and gamma that yields the same solution.**

Equating the two Euler conditions,

$$U_{CB} - U_n = \lambda\beta(\pi_{CB} - \pi^*) \qquad \text{and} \qquad (U_{CB} - U_n)^3 = \gamma\beta(\pi_{CB} - \pi^*)^3$$

we obtain the equation explaining the relationship between these two weights.

$$\gamma^{\frac{1}{3}}\beta^{\frac{1}{3}}(\pi_{CB} - \pi^*) = \lambda\beta(\pi_{CB} - \pi^*)$$

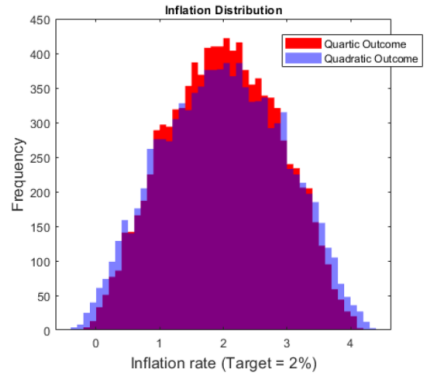
$$\boxed{\gamma = \lambda^3\beta^2}$$

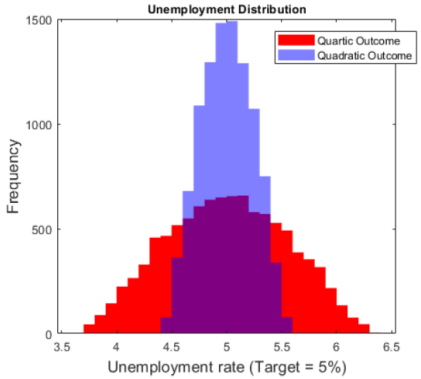
Structural Parameter:
Cost of lowering inflation

Remove excess penalization
in quartic form

Summary: Properties of Quartic Loss Function

1. Monte Carlo Simulation

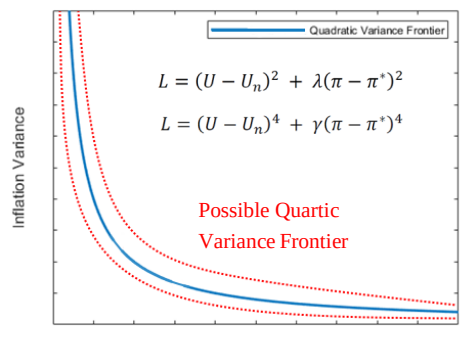


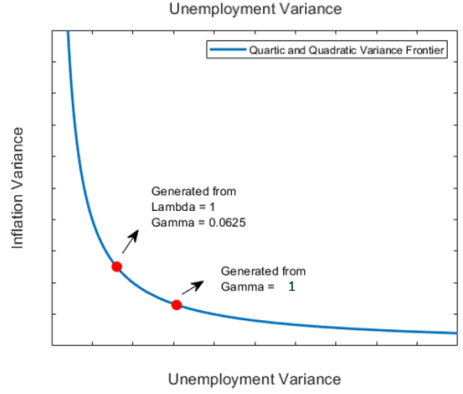


Source: Author's calculation

Span

2. Taylor Curve





Source: Author's calculation

Write

Transform

3. Gamma Transformation

Since these conditions yield the same variance,

$$(U_{CB} - U_n)^3 = \gamma\beta(\pi_{CB} - \pi^*)^3$$

$$U_{CB} - U_n = \lambda\beta(\pi_{CB} - \pi^*)$$

Hence, they generate the same set of solutions.

Equating two Euler conditions,

$$\gamma^{\frac{1}{3}}\beta^{\frac{1}{3}}(\pi_{CB} - \pi^*) = \lambda\beta(\pi_{CB} - \pi^*)$$

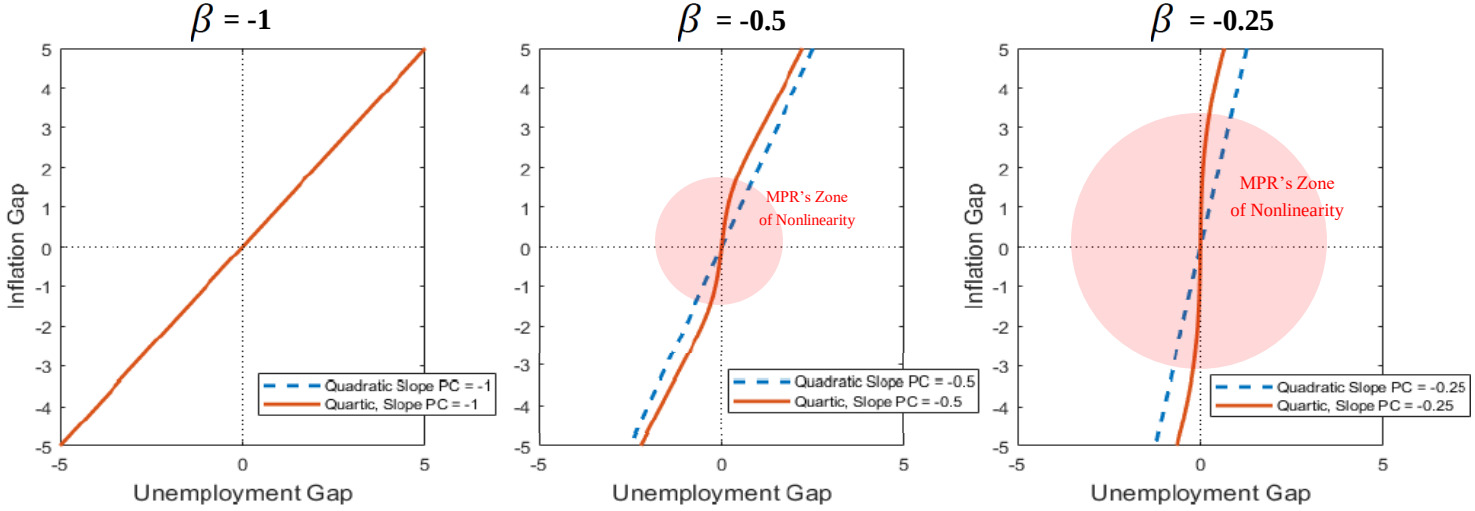
$$\boxed{\gamma = \lambda^3\beta^2}$$

Source: Author's Derivation

What exactly does the quartic tell us ?

Higher Moments, Highly Matter: Uncertainty x Structure of the Economy

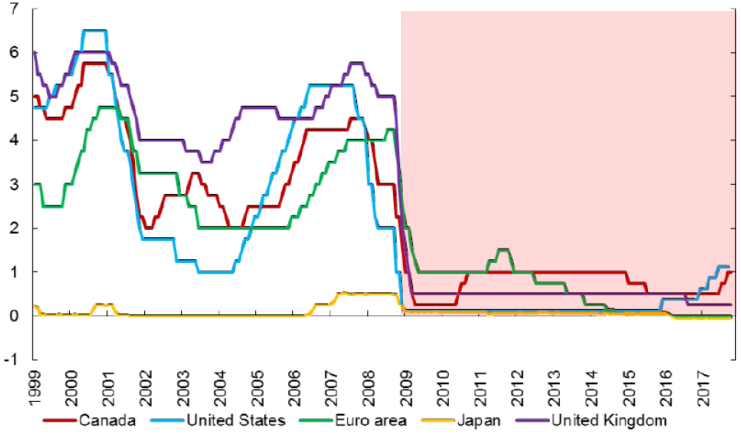
- In case the cost of reducing inflation becomes larger, it pay for the policy makers to wait and see (zone of nonlinearity expands), since they are subjected to large demand shock and now lowering inflation greater exposes the economy to large deviation in unemployment.
- This implies that the higher moments of shocks in fact highly matter as **they have the duality of distributional and structural implications for the economy.**



Source: Author's calculation

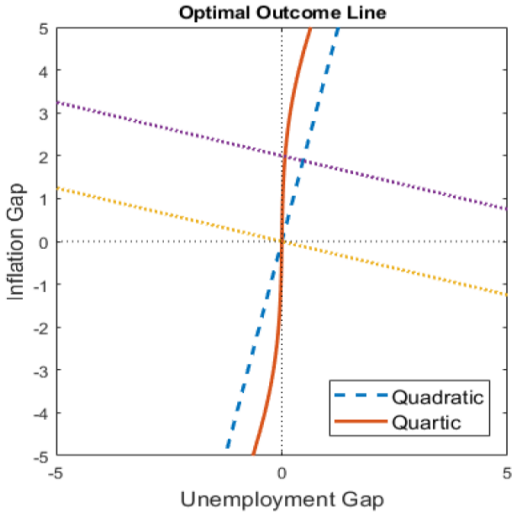
Some Evidence on Risk Management

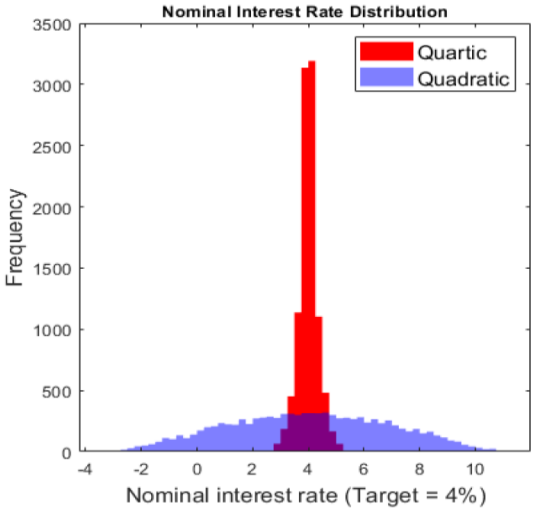
- As an example, consider the time when the economy is surrounded by large uncertainty on output, say, the global financial crisis or even the global pandemic, the central bankers try to minimize variation in output gap given the knowledge of possible large second round demand shock.
- **In presence of such uncertainty, closing output gap, rather than inflation gap, is the top priority.**



Monetary Policy Rate, monthly (%)

Source: Mendes Murchisons and Wilkins (2017)





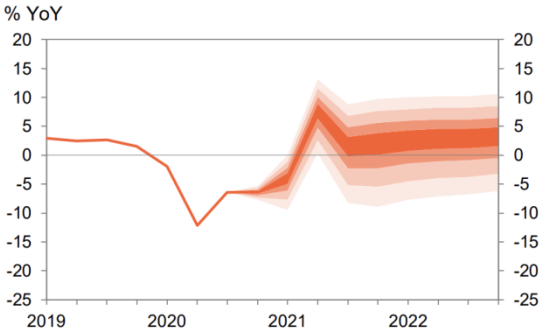
Source: Author's calculation

How can the quartic loss function be employed in CBs?

- One may argue that the quartic loss function is hard to be solved, especially in more complex models.
- But in a lager sense, the essence of this functional form is its capacity to take into account the shape of uncertainty at any given point of time (moments of risk).
- The policy question is to what extent should we incorporate uncertainty into our decision. Hence if we are to build some models, maybe even the simplest structure can answer that question.

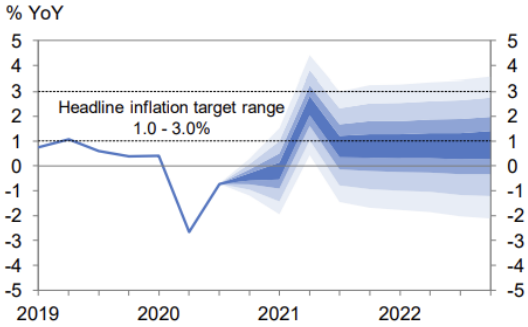
- **Just figure out the variances and skewness of shocks to GDP and inflation!** One possibility is that we can model time varying distribution based on ar(p) structure, that is easier than predicting exactly what will happen in the future.

Chart 2.22 Growth forecast



Note: Fan chart covers 90% of the probability distribution

Chart 2.23 Headline inflation forecast



Note: Fan chart covers 90% of the probability distribution

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Concluding Remarks

- To my knowledge, this paper is **among the first** in examining theoretically how monetary policy should respond to tail risk.
- Changing economic structure can create nonlinearity in optimal policy response.
  - In quadratic case, changing beta just alters the slope of response but not the shape as in the quartic case.
  - This implies that standard Taylor rule is not well suited.
- The big question: What is the future of monetary policy in a rapidly changing world?
  - This paper gears toward "risk-managed" monetary policy.

Food for Thought:

Should monetary policy focus less on stabilization but more on preventing tail risk?

Thank You

Q & A Session